



Study on temperature-controlled by phase change materials and energy analysis in a refrigerated vehicle during the door opening

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ARTICLE INFO

Keywords:

Refrigerated vehicles
Phase change materials
Door opening
Temperature controlled
Energy analysis
Numerical simulation

ABSTRACT

This study investigates how phase change materials (PCMs) mitigate temperature rise and energy consumption during door openings in refrigerated vehicles at 66 % loading capacity. Numerical simulations are employed to analyze airflow patterns, thermal dynamics, and energy consumption. Key findings include: When the fan is off, external hot air enters the compartment, raising internal temperatures and impacting both the food and the PCM. With the fan on, airflow reduces direct contact between the hot air and the food, and increasing inlet velocity helps prevent excessive temperature rise. However, an inlet velocity of 3 m/s causes the air to escape the compartment, diminishing the cooling capacity of the PCM and raising the internal temperature. Baffle designs effectively block hot air, particularly in areas near the door, mitigating temperature rise and slowing food warming, thus reducing recovery energy. Higher inlet velocities increase cold energy consumption, with an inlet velocity of 3 m/s showing the highest consumption due to direct airflow from the fan. However, baffles reduce cold energy usage. The inlet velocity with 2 m/s in the baffle case is identified as optimal, balancing temperature control, food quality preservation, and lower recovery energy consumption. It also improves cold energy efficiency and reduces hot air intrusion.

1. Introduction

With the rapid development of global cold chain logistics, refrigerated transportation plays a crucial role in ensuring the quality and safety of goods such as food and pharmaceuticals [1,2]. Refrigerated vehicles, as the core equipment in cold chain logistics, directly impact transportation efficiency and costs. However, refrigerated transportation vehicles face numerous challenges during operation, including controlling energy consumption during long-distance transport, increased heat load caused by frequent door openings, and the stability of vehicle insulation performance under different environmental conditions [3].

Researchers have extensively studied energy consumption and carbon dioxide emissions in refrigerated vehicles. Adekomaya et al. found that using aluminum plates as the outer walls of refrigerated vehicles shortens the lifespan of insulation materials and exacerbates heat penetration, leading to increased energy consumption [4]. Maiorino et al. applied electric energy recovery systems in light-duty refrigerated vehicles, providing an innovative solution for low-carbon cold chain

transportation [5]. Fabris et al. successfully reduced fuel consumption growth from 19.0 % to 5.7 % and significantly lowered pollutant emissions through battery-driven compressor technology [6]. Song et al. optimized the refrigeration system in electric refrigerated vehicles, achieving an average energy consumption reduction of 0.5 kW and an improved COP value, extending battery range [7]. Maiorino et al.'s hybrid refrigerated vehicle integrated photovoltaic panels with batteries, significantly reducing engine greenhouse gas emissions [8]. Meneghetti et al. installed solar panels on the rooftops of refrigerated vehicle, combined with battery packs and power conversion devices, replacing traditional diesel refrigeration units and reducing pollution emissions [9]. Yao et al. developed a waste heat recovery system that performed excellently under various road conditions, providing 10 kW of fuel-free cooling capacity, achieving a 6.3 % reduction in carbon emissions [10]. Gao et al. innovatively applied MnCl₂/CaCl₂-NH₃ working fluids in thermochemical adsorption-based energy storage refrigeration systems for refrigerated vehicles, achieving an energy efficiency of 1.4 at -20 °C, doubling the performance of conventional

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<https://doi.org/10.1016/j.icheatmasstransfer.2025.109824>

Table 1
Insulation material and interior dimension of the refrigerated vehicle.

Insulation material					Interior dimension		
Material	Thickness- mm	Density- kg/m ³	Specific heat capacity- J/(kg·K)	Thermal conductivity- W/(m·K)	Width	Height	Depth
Polyurethane	50	50	1470	0.022	1.6	1.7	3.0

systems [11]. Pan et al. proposed a supercritical CO₂ cycle and dual-temperature refrigeration cycle system, demonstrating its potential for energy savings and emissions reduction through thermodynamic and economic evaluations [12]. Li et al. introduced a new absorption refrigeration system driven by the waste heat of proton exchange membrane fuel cells, improving the refrigerated vehicle's range while reducing operating costs [13].

During refrigerated transportation, the increased heat load caused by door openings is a prominent issue. Every door opening introduces a large amount of external warm air, significantly increasing the vehicle's cooling demand and energy consumption [14]. This problem is particularly severe in multi-stop refrigerated delivery vehicles, where heat infiltration is especially problematic. Therefore, a deep understanding of airflow and heat transfer characteristics during door opening processes and their influencing factors is crucial for optimizing the energy consumption and performance of refrigerated vehicles.

Significant progress has been made in related research in this field. Zhang et al. applied computational fluid dynamics (CFD) technology to systematically study the impact of different door opening sizes and positions on the internal temperature distribution [15]. Rai et al. explored the mechanism of air curtains in inhibiting heat infiltration during door opening in refrigerated vehicles and confirmed that an air curtain with a wind speed of 3.1 m/s could reduce energy consumption by nearly 48 % [16]. Luchini et al. applied model predictive control algorithms to achieve precise temperature regulation and effectively addressed the energy consumption impact of door openings [17]. Jeong et al. found through experiments that setting cold plates at the top of refrigerated vehicles could significantly reduce the internal temperature during door openings in an empty load state [18].

Phase change materials (PCMs), with significant energy-saving potential, have been widely researched and applied in cold chain transportation [19–24]. Ahmed et al. modified the insulation structure of traditional refrigerated trailers using PCM, and experimental results showed that PCM significantly reduced the daily average heat infiltration into the vehicle, reducing the heat load [25]. Calati et al. developed polyurethane-based composite insulated walls by embedding 0.5–2 cm thick PCM layers into traditional insulation layers, and even the thinnest PCM layers maintained stable temperature during the simulation period [26]. Liu et al. proposed a new phase change cold storage unit technology, which, when applied in refrigerated vehicles, significantly improved the compartment's temperature control performance and achieved energy-saving effects [27]. Ben Taher et al. validated the PCM integration technology, demonstrating that the PCM system could significantly reduce refrigeration power demand when the door was opened, thereby reducing emissions and saving energy [28]. Kang et al. studied the impact of the types, shape, and amount of PCMs on the quality of cold chain transportation, providing theoretical support for the engineering application of PCM [29].

However, the research gap addressed by this study is the lack of consideration for cargo loading when assessing the impact of PCMs on temperature rise and energy consumption in refrigerated vehicles. This study fills that gap by exploring PCM's impact on temperature rise and energy consumption during door openings in refrigerated vehicles with a 66 % loading rate, providing a theoretical foundation for optimizing refrigerated vehicle performance under practical conditions.

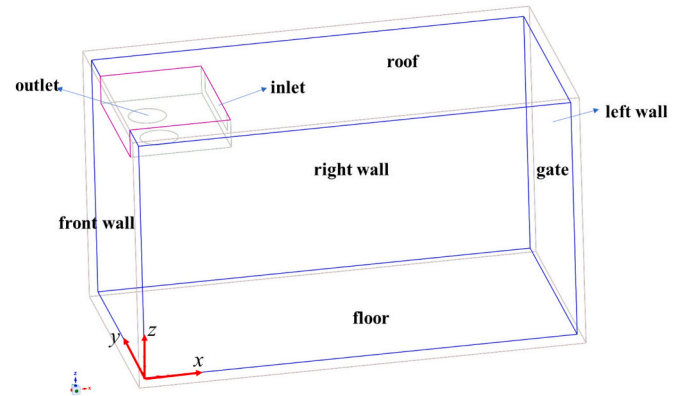


Fig. 1. Geometry and boundaries of the refrigerated vehicle.

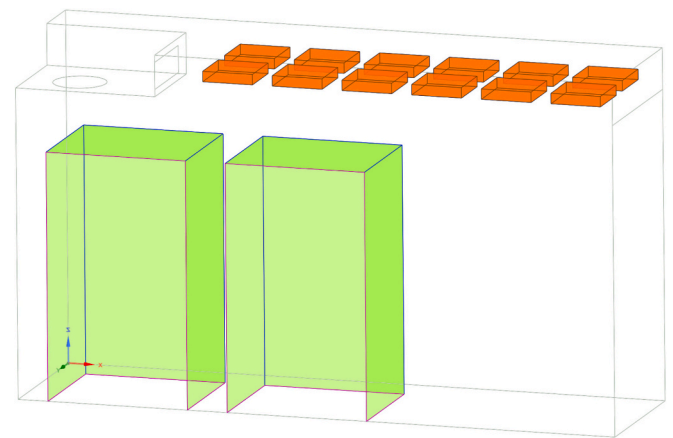


Fig. 2. Schematic PCM-based cooling system.

2. Description of the study cases

2.1. Refrigerated vehicle

The structure of the refrigerated vehicle is composed of thermal insulation sandwich plate [30,31]. The thickness of the compartment plate is 51.7 mm and the expanded polyurethane foam is used as the central insulation material. The thermal parameters of the polyurethane and the interior dimensions of the refrigerated vehicle are shown in Table 1. The geometry and boundary conditions of the refrigerated vehicle are shown in Fig. 1.

2.2. PCM-based cooling scheme

To lower the fuel consumption, this paper proposes a PCM-based cooling system for refrigerated vehicles that does not require an on-board refrigeration system, as shown in Fig. 2. The PCM is cooled to a certain low temperature at the cooling station before being placed in the device, which is installed on the top of the refrigerated vehicle. Once the fan is activated, air is blown over the PCM, where it is cooled before entering the return air grilles (outlet). After circulating through the fan,

Table 2
Parameters of the PCM and food.

Materials	Dimensions (depth × width × thickness)-mm	Number	Arrangement	Height- mm	Spacing- mm
RTO (PCM)	250 × 250 × 50	24	inline	1600	100
Food	700 × 1200 × 1200	2	inline	600	200

the cooled air is sent back into the vehicle compartment, where it exchanges heat with the PCM and is further cooled. This cycle continues, and the air inside the vehicle compartment will be continuously cooled until thermal equilibrium is reached, maintaining a specific low temperature. Two shelves of food are loaded in the compartment, as shown in Fig. 2. The parameters and the thermo-physical characteristics of the PCM and the food are displayed in Tables 2 and 3, respectively.

This proposed solution not only eliminates the need for a refrigeration system but also enables the use of inexpensive off-peak electricity to charge the PCM, thereby further reducing the operating costs of refrigerated vehicles. Additionally, based on the transportation time of refrigerated vehicles, the number of the PCM units can be increased or decreased, thereby enhancing the flexibility of the PCM-based cooling system.

During door openings, a significant amount of hot air from outside enters the compartment. The hot air increases the temperature of both the food and the air in the compartment. For the PCM-based cooling scheme, the effects of the inlet velocity and the baffle at the door are studied. The baffle, with a height of 200 mm, is located at the top of the door, as shown in Fig. 2. The function of the baffle is to prevent the cold air from blowing into the outside environment. The inlet velocities of the door opening cases with or without a baffle are listed in Table 4. The range of the inlet velocity is based on the design specifications of the actual refrigerated vehicle's fan.

2.3. Numerical method

2.3.1. Equations

To simplify the calculation, the following reasonable assumptions are made. Air is a Newtonian fluid and incompressible. The thermal properties of air are temperature-dependent. Considering gravity effect, the effect of buoyancy is approximated by Boussinesq hypothesis.

Based on these assumptions, the mass, momentum, and energy conservation equations are given below:

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0 \quad (1)$$

$$\frac{\partial \bar{u}_i}{\partial t} + \frac{\partial (\bar{u}_i \bar{u}_j)}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \nu \frac{\partial^2 \bar{u}_i}{\partial x_j^2} + \frac{\partial (-\bar{u}_i \bar{u}_j)}{\partial x_j} + \delta_{3i} g_i \beta_t (\bar{T} - T_0) \quad (2)$$

$$\frac{\partial \bar{T}}{\partial t} + \frac{\partial (\bar{u}_i \bar{T})}{\partial x_i} = \frac{\partial}{\partial x_i} \left[\left(\frac{\nu}{Pr} + \frac{\nu_t}{Pr_t} \right) \frac{\partial \bar{T}}{\partial x_i} \right] \quad (3)$$

The relationship between Reynolds stress and turbulent viscosity is as follows:

$$-\bar{u'_i u'_j} = 2\nu_t S_{ij} + \frac{2}{3} k \delta_{ij} \quad (4)$$

Table 3
Thermo-physical characteristics of the PCM and food.

Material	Density-kg/m ³	Specific heat-J/(kg·K)	Thermal conductivity-W/(m·K)	Latent heat of fusion-kJ/kg	Solidus temperature-°C	Liquidus temperature-°C
RTO(PCM)	Solid: 880 Liquid: 770	Solid: 2000 Liquid: 2000	Solid: 0.2 Liquid: 0.2	175	-1	2
Food	300	1000	0.2	-	-	-

Solving for turbulent viscosity in Eq. (4) is the main challenge of the RANS method. This paper adopts the widely used BSL $k-\omega$ model [32]. The efficiency of this model has been proven to be higher than that of other models under the same accuracy in our previous paper [33].

The equation for the BSL $k-\omega$ model is as follows:

$$\frac{\partial k}{\partial t} + \frac{\partial}{\partial x_j} (\bar{u}_j k) = P_k - Y_k + \frac{\partial}{\partial x_j} \left[\left(\nu + \frac{\nu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] \quad (5)$$

$$\frac{\partial \omega}{\partial t} + \frac{\partial}{\partial x_j} (\bar{u}_j \omega) = \alpha \frac{\omega}{k} P_k - Y_\omega + \frac{\partial}{\partial x_j} \left[\left(\nu + \frac{\nu_t}{\sigma_\omega} \right) \frac{\partial \omega}{\partial x_j} \right] + 2(1-F) \frac{1}{\sigma_{\omega 2} \omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j} \quad (6)$$

where F is the blending function, whose formulas is as below:

$$F = \tanh(\Phi^4) \quad (7)$$

$$\Phi = \min \left[\max \left(\frac{\sqrt{k}}{0.09 \omega d_w}, \frac{500 \nu}{d_w^2 \omega} \right), \frac{4 \rho k}{\sigma_{\omega 2} D_\omega d_w^2} \right] \quad (8)$$

$$D_\omega = \max \left(2 \frac{\rho}{\sigma_{\omega 2} \omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j}, 10^{-10} \right) \quad (9)$$

The model parameters are computed as follows:

$$\sigma_k = \frac{1}{F/\sigma_{k1} + (1-F)/\sigma_{k2}} \quad (10)$$

$$\sigma_\omega = \frac{1}{F/\sigma_{\omega 1} + (1-F)/\sigma_{\omega 2}} \quad (11)$$

$$\beta = F\beta_1 + (1-F)\beta_2 \quad (12)$$

$$\alpha = F\alpha_1 + (1-F)\alpha_2 \quad (13)$$

The solidification/melting model of PCMs adopts the enthalpy-porosity method [34] to simulate the phase change process. This method does not track the two-phase interface and introduces a quantity called the liquid fraction, which represents the volume fraction of the liquid phase in each computational cell. There exists a mushy zone between the two phases, where the liquid fraction ranges from 0 to 1. The mushy zone is modeled as a 'pseudo' porous medium, where the porosity decreases from 1 to 0 as the material solidifies. The mushy zone constant is 100,000. The enthalpy-porosity method is particularly advantageous because it simplifies the complex process of phase change modeling, improves computational efficiency, and can be applied in a variety of systems and geometries, making it an essential tool in many engineering and scientific applications. The enthalpy-porosity method has been widely validated and applied in literatures [35,36]. In this method, the enthalpy of the PCM is the sum of sensible enthalpy and latent heat

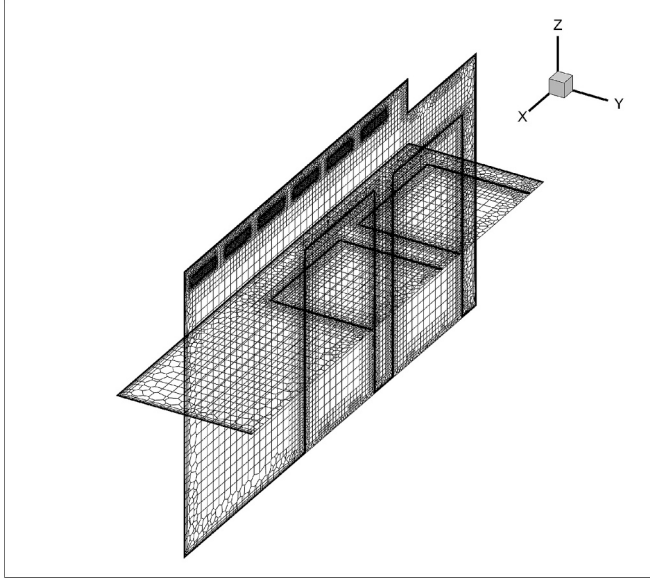
Table 4
Inlet velocity of the door opening cases with or without a baffle.

Cases	Baffle	Inlet velocity-m/s	Re	Gr
Case-V0	No	0	-	1.7e10
Case-V1	No	1	2.6e4	-
Case-V2	No	2	5.2e4	-
Case-V3	No	3	7.8e4	-
Case-V2-baffle	Yes	2	5.2e4	-
Case-V3-baffle	Yes	3	7.8e4	-

Table 5

Initial conditions and Boundary conditions for the PCM-based cooling case.

Case	Initial conditions		Boundary conditions		Outlet	Exterior surface	
	Velocity- m/s	Temperature- °C	Inlet	Temperature-°C		Flow	Thermal
PCM-based cooling case	0	PCM: -1 Air: 4 Food: 4	Table 4	Equal to the time-averaged temperature at the outlet from the previous time step	Outflow	Non-slip wall	$h_e = 3.41 \text{ W/(m}^2\text{K)}$, outside air temperature = 22.5 °C

**Fig. 3.** Grid layout.

content, and the calculation formula is as follows:

$$H = h + \Delta H \quad (14)$$

$$h = h_{ref} + \int_{T_{ref}}^T c_{p-PCM} dT \quad (15)$$

The liquid fraction β_{lf} is calculated as:

$$\begin{aligned} \beta_{lf} &= 0 \quad \text{if } T < T_s \\ \beta_{lf} &= 1 \quad \text{if } T > T_l \\ \beta_{lf} &= \frac{T - T_s}{T_l - T_s} \quad \text{if } T_s < T < T_l \end{aligned} \quad (16)$$

ΔH is calculated as:

$$\Delta H = \beta_{lf} L \quad (17)$$

where L is the latent heat.

The energy equation of the PCM is as below.

$$\frac{\partial H}{\partial t} + \frac{\partial}{\partial x_i} (\bar{v}H) = \frac{\partial}{\partial x_i} \left[\frac{\lambda}{\rho_{PCM}} \frac{\partial T}{\partial x_i} \right] \quad (18)$$

2.3.2. Initial and boundary conditions

The initial and boundary conditions for the validation case and the PCM-based cooling case are shown in Table 5.

2.3.3. Numerical schemes

The numerical scheme is outlined below:

Table 6

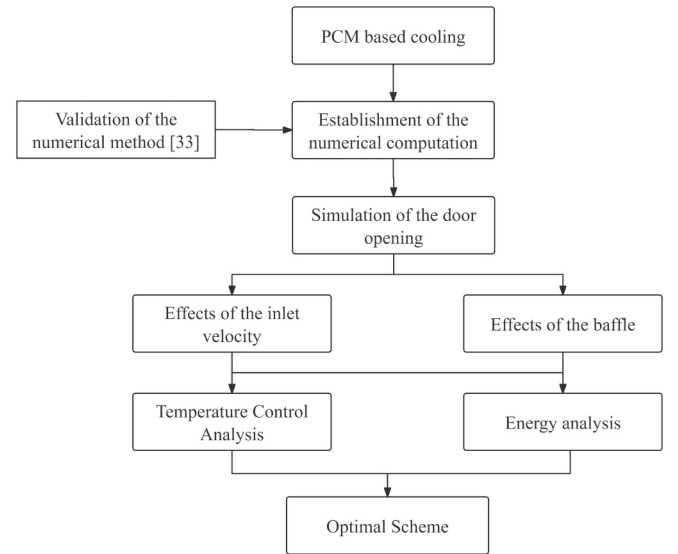
Grid resolution.

Grids for boundary layer			Size of core grids-mm	Size of face grids-mm		Grid number/ten thousand
Size of first grid-mm	Growth rate	Number of layers		inlet, outlet	others	
1	1.2	10	50	15	45	26

Table 7

Convergence criterion and simulation time.

Convergence criterion		Simulation time-s	Time step-s
Energy equation	Other equations		
10e-8	10e-5	600	0.05

**Fig. 4.** Flowchart of the study.

- (1) The time term is discretized using the first order implicit scheme.
- (2) The convection term adopts the second-order upwind format.
- (3) The diffusion term adopts the central difference scheme.
- (4) The velocity and pressure coupling method is implemented by SIMPLEC method.

2.3.4. Grids, convergence criterion and simulation time

The grid layout is shown in Fig. 3. The grid near the walls is polyhedral cells, and the center area is uniform hexahedral cells. The specific grid size is shown in Table 6. For the PCM-based cooling case, due to the same outside air temperature, half of the computing domain is considered.

The convergence criterion and simulation time are shown in Table 7.

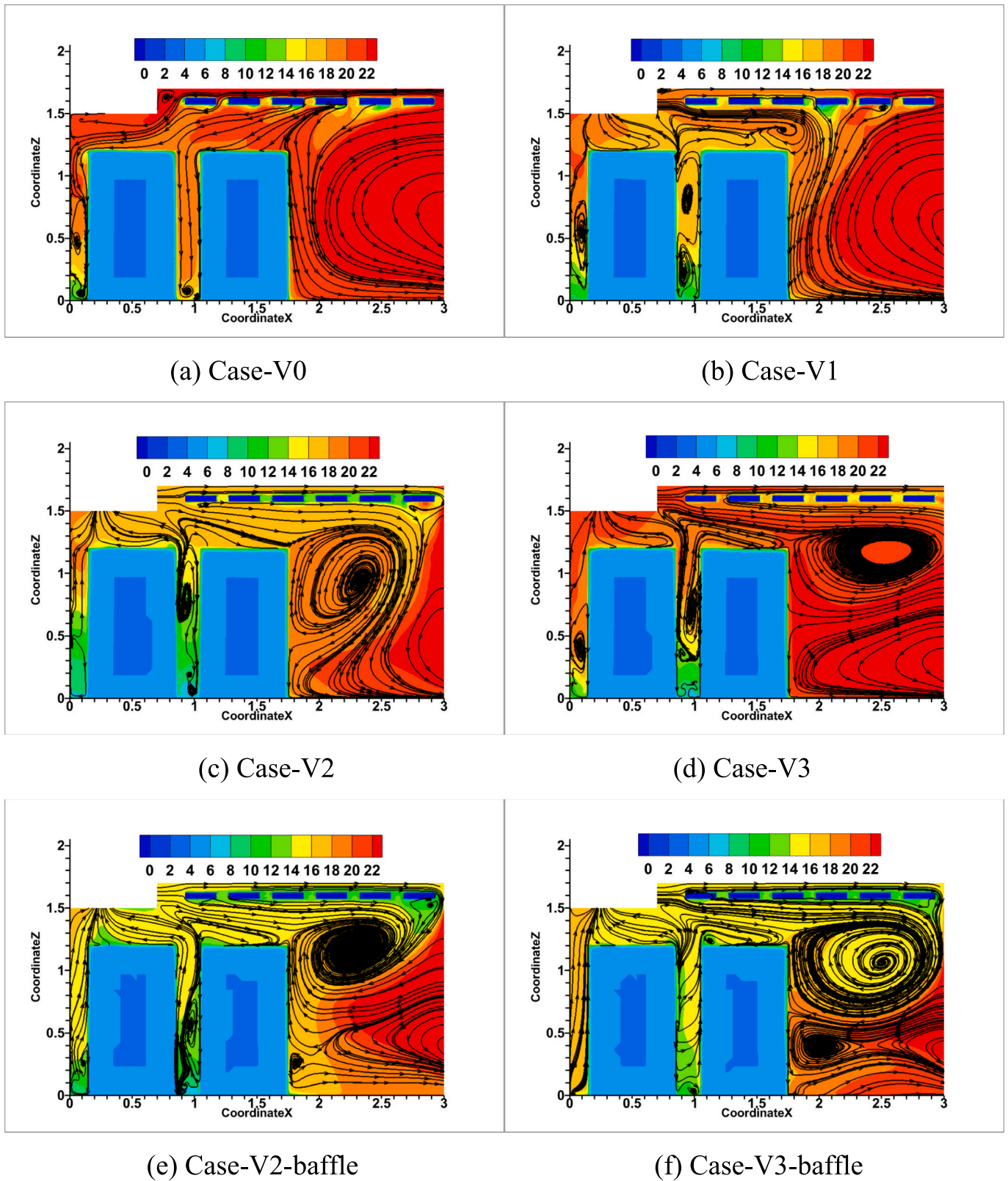


Fig. 5. Streamlines and temperature contours on the plane $y = 0.6$ m at 600 s.

The used numerical method has been validated in our previous paper [33]. And the Ansys Fluent Student 2022 was used for simulating. And, the flowchart of the study is shown in Fig. 4.

3. Results and discussion

3.1. Air flow and temperature

Fig. 5 shows the streamlines and temperature contours at the $y = 0.6$

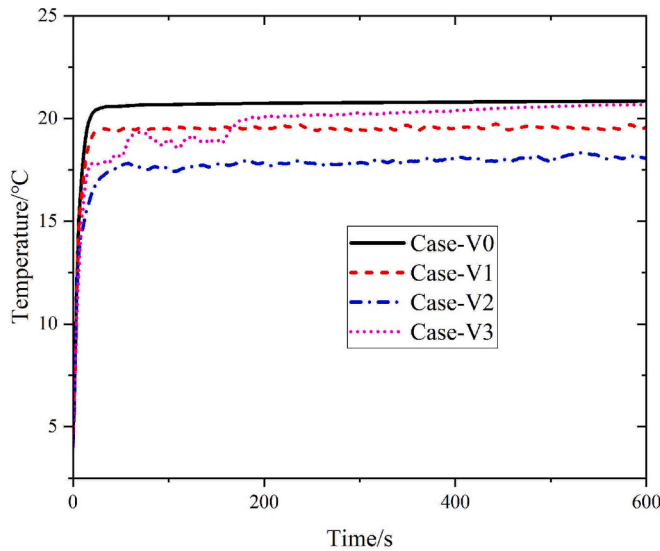


Fig. 6. Time histories of the air temperature without the baffle.

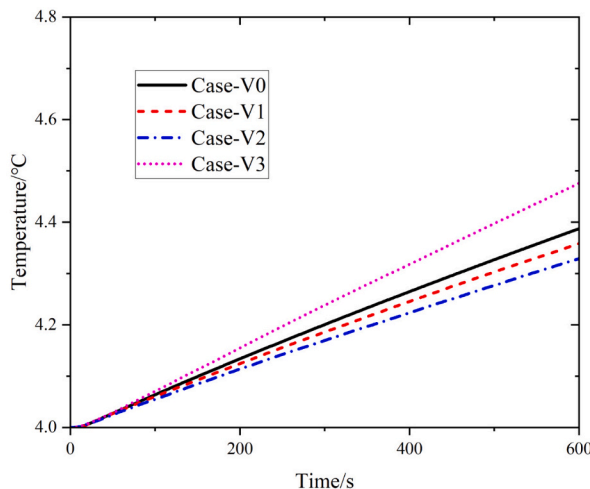
m plane at 600 s. Fig. 5-a demonstrates the scenario when the fan is off. Due to the higher external air temperature compared to the internal air temperature, the external air flows into the compartment from the upper part of the door and exits from the lower part of the door. The hot air entering the compartment exchanges heat with the PCM and the food, leading to a decrease in the temperature. However, the temperatures of the PCM and the food increase. Fig. 5-b and -c illustrates the scenarios when the fan is activated. The air from the fan counteracts the external hot air in the upper part of the compartment. This effectively prevents some of the hot air from coming into contact with the PCM and the food, and exchanging heat. When the inlet velocity is 2 m/s, an eddy forms near Food-2. The low inlet-velocity air effectively reduces the contact of hot air with the food, keeping the air temperature between the food and the door lower. However, when the inlet velocity reaches 3 m/s, the air velocity is too high, causing the air to directly exit the compartment, resulting in a higher internal air temperature than in other scenarios. To prevent this, a baffle is installed at the top of the door to block the air from exiting, thereby reducing cold loss. Fig. 5-e and -f shows that the baffle effectively blocks the air from the fan, preventing hot air from

directly contacting the food. When the air from the fan encounters the external air, eddies form in the areas where no food is loaded. As the inlet velocity increases, the size of the eddies also increases. The temperature field shows that the baffle effectively lowers the air temperature within the compartment, with high-temperature regions primarily concentrated near the door.

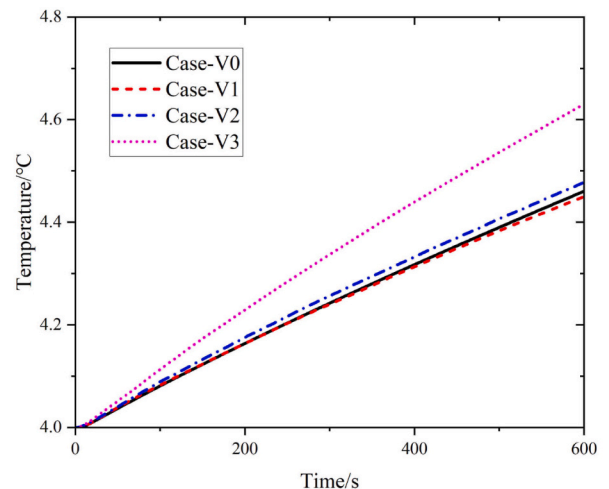
Fig. 6 shows the time histories of the air temperature without the baffle. It can be observed that the entry of external air causes the air temperature to rise rapidly in a short period and then slowly increase. When the inlet velocity is below 3 m/s, the air temperature decreases as the inlet velocity rises. The higher the inlet velocity, the better the effect of blocking external air and the better the heat exchange between the air from the fan and the PCM, resulting in more cooling. However, when the inlet velocity is 3 m/s, the air from the fan is directly discharged from the compartment, and the cooling effect is only significant during the initial opening of the door. As time progresses, external hot air continuously enters, and the cooling effect of the air from the fan gradually weakens, ultimately bringing the internal air temperature close to that observed when the fan is deactivated.

Fig. 7 shows the time histories of the food temperature (volume averaged) without the baffle. Food-1 is positioned near the front wall of the compartment, while Food-2 is at the center. When the inlet velocity is below 3 m/s, the temperature of Food-1 decreases as the inlet velocity increases, while the temperature increase of Food-2 is not significantly affected by the inlet velocity. As shown in Fig. 5 a-c, the air temperature near Food-1 decreases as the inlet velocity increases. This is because the air from the fan exchanges heat with the PCM, and the greater the inlet velocity, the larger the heat exchange, resulting in a lower air temperature. The air temperature near Food-2 is lower, but the air velocity near Food-2 increases as the inlet velocity increases, leading to more heat being transferred to Food-2. Therefore, when the inlet velocity is 2 m/s, the temperature near Food-2 increases. When the inlet velocity is 3 m/s, the air from the fan directly carries away cooling energy from the PCMs, resulting in the greatest temperature increase for the food. Therefore, the inlet velocity should be controlled below 3 m/s, or the temperature rise of both the food and the internal air will become more severe.

Fig. 8 shows the time histories of the liquid fraction (volume averaged) of the PCM without the baffle. As the inlet velocity increases, the heat exchange efficiency between the air and the PCM improves, and thus the melting rate of the PCM and its liquid fraction increase with higher inlet velocity.



(a) Food-1



(b) Food-2

Fig. 7. Time histories of the food temperatures without the baffle.

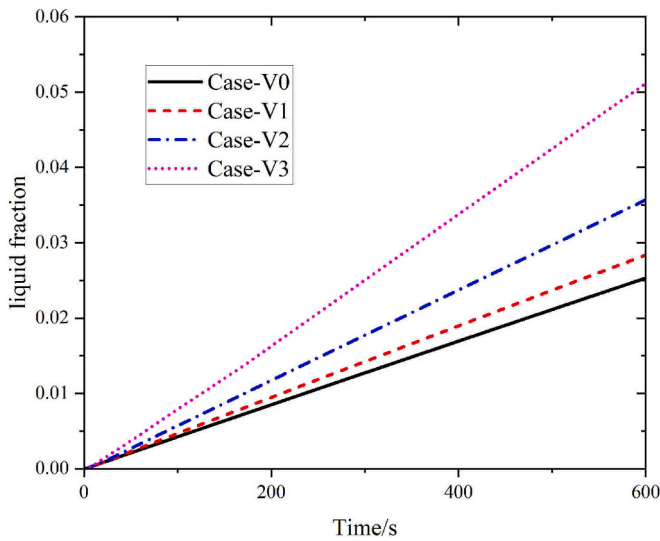


Fig. 8. Time histories of the liquid fraction of the PCM without the baffle.

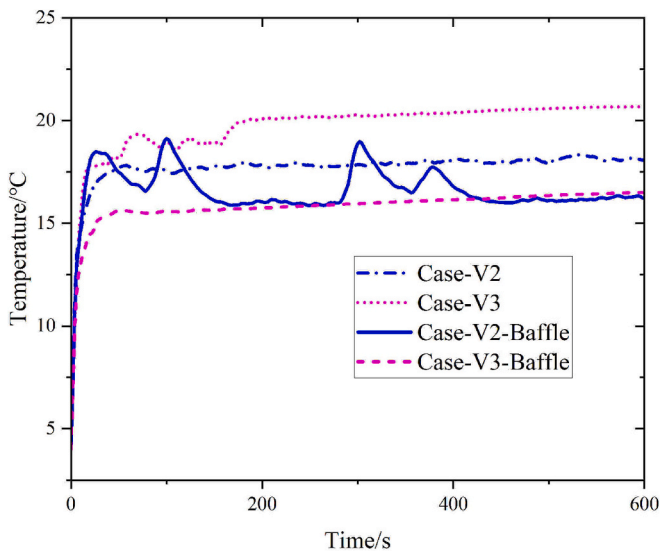


Fig. 9. Time histories of the air temperature with or without the baffle.

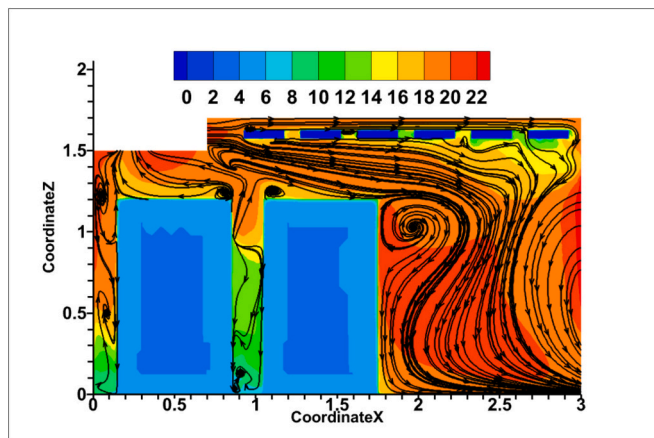


Fig. 10. Streamlines and contours of temperature on the plane $y = 0.6$ m for Case-V2-baffle at 300 s.

Fig. 9 shows the time histories of the air temperature (volume averaged) with or without the baffle. The temperature at the end of the door opening indicates that the baffle can effectively lower the air temperature, especially when the inlet velocity is 3 m/s. Notably, the air temperature of Case-V2-baffle is lower than that of Case-V2, but during the entire door opening process, the air in Case-V2-baffle experiences periods of higher temperature. This occurs because the direction of the fan airflow deviates during the flow process, reducing the heat exchange efficiency between the air and the PCM, thereby creating more high-temperature regions. This can be observed from Fig. 10 and Fig. 5-e.

Fig. 11 shows the time histories of the food temperature (volume averaged) with or without the baffle. It can be seen that the baffle has a small effect on the temperature rise of the food, except for Food-2 in Case-V3-Baffle and Case-V3. The baffle effectively reduces the temperature rise of Food-2 when the inlet velocity is 3 m/s. The temperature rise of Food-1 is less affected, as Food-1 is shielded by Food-2 and thus is less impacted by the airflow.

Fig. 12 shows the time histories of the liquid fraction (volume averaged) of the PCM with or without the baffle. Over time, the liquid fraction gradually increases. The liquid fraction growth rate in Case-V2-Baffle and Case-V3-Baffle, is slower than in the corresponding cases without baffles (Case-V2 and Case-V3), indicating that the baffle design effectively hinders cold energy loss.

3.2. Energy analysis

This section provides an analysis of the energy aspects, which include both the recovery energy and the cold energy consumption of the PCM. The recovery energy is defined as the energy required to restore the temperature of the internal air and food to their initial set temperature. This recovery energy serves as a key metric for evaluating the extent of warm air infiltration and is determined through the following calculation:

$$E_r = \int_V \rho C_p (T - T_0) dV \quad (19)$$

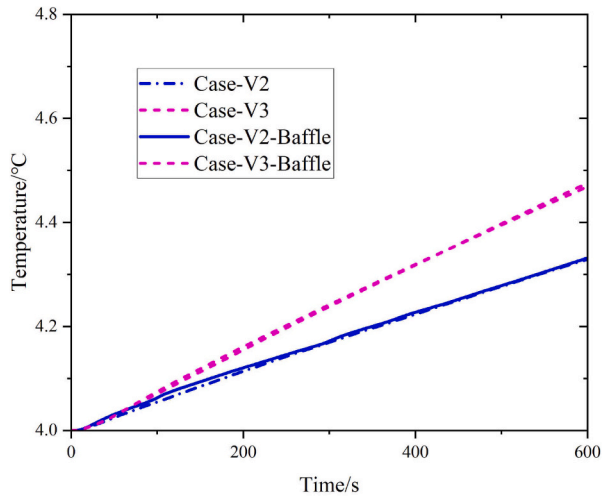
As shown in Table 8, in all cases, Case-V3 has the highest total recovery energy, amounting to 225.20 kJ, significantly higher than the other cases, especially Case-V2 (170.88 kJ) and Case-V2-baffle (162.52 kJ). Case-V3 not only has the highest total recovery energy but also the highest individual recovery energy. This is because, in Case-V3, the air from the fan directly carries away cooling energy from the PCM, resulting in the maximum recovery energy for both the air and food. Case-V2-baffle has the smallest recovery energy at 162.52 kJ because the baffle effectively obstructs the air from taking away the cooling energy, lowering the temperature around Food-2 and reducing the total recovery energy.

The recovery energy of the air is relatively low, typically between 40 and 60 kJ, because air has a small heat capacity and the internal air is in contact with the PCM. The recovery energy of Food-1 and Food-2 varies across different cases, especially Food-2, which has higher recovery energy in all cases, particularly in Case-V3 (95.27 kJ). This is because Food-2 is near the door and is more influenced by the hot air, resulting in a higher temperature rise.

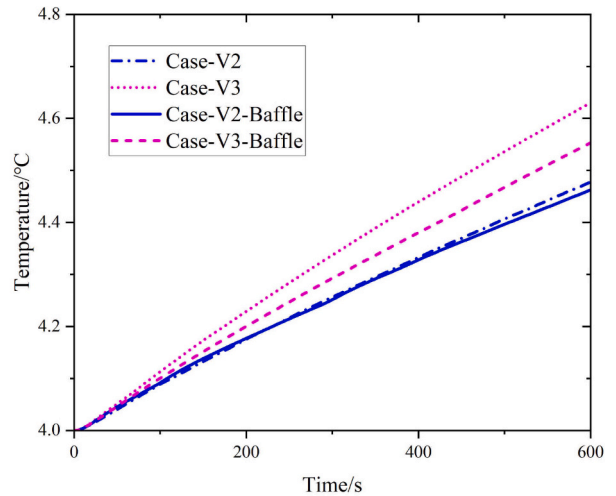
The cold energy consumption of the PCM includes two components: sensible heat and latent heat. Its calculation formula is as follows:

$$E_c = \int_V \rho C_p (T - T_0) dV + \int_V \rho (\beta_{li} - \beta_{li0}) L dV \quad (20)$$

Table 9 lists the cold energy consumption of the PCM in different cases. From Case-V1 to Case-V3, cold energy consumption increases significantly, especially in Case-V3, where the cold energy consumption is notably higher than that in other cases. This is due to the air from the fan directly carrying the PCM's cooling energy out of the compartment, and the maximum velocity increasing the heat exchange efficiency. In



(a) Food-1



(b) Food-2

Fig. 11. Time histories of the food temperatures with or without the baffle.

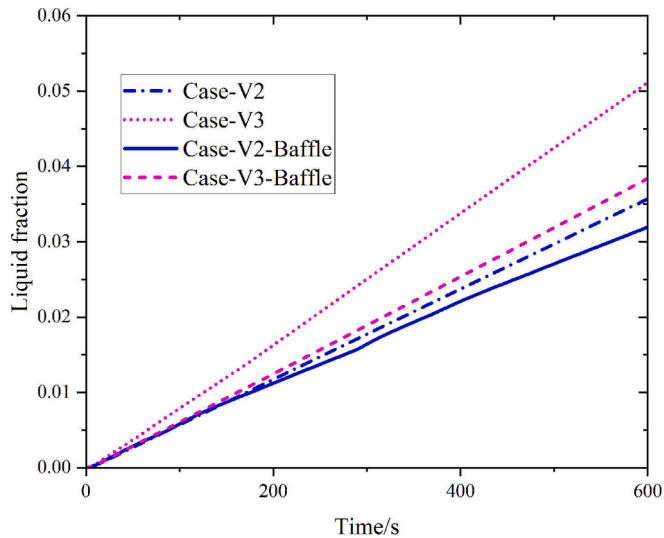


Fig. 12. Time histories of the liquid fraction of the PCM with or without the baffle.

Table 8
Recovery energy.

Cases	Recovery energy-kJ/%			
	Air	Food1	Food2	Sum
Case-V0	58.58/31.36 %	58.59/31.37 %	69.60/37.27 %	186.77
Case-V1	54.16/30.70 %	54.19/30.72 %	68.03/38.57 %	176.38
Case-V2	48.89/28.61 %	49.77/29.13 %	72.22/42.26 %	170.88
Case-V3	57.99/25.75 %	71.94/31.94 %	95.27/42.30 %	225.20
Case-V2-baffle	42.44/26.11 %	50.15/30.86 %	69.93/43.03 %	162.52
Case-V3-baffle	43.48/21.96 %	70.87/35.80 %	83.62/42.24 %	197.97

cases with baffles (Case-V2-baffle and Case-V3-baffle), cold energy consumption is reduced by 10.5 % and 25.0 %, respectively, compared to the corresponding cases without baffles, indicating that the baffle design effectively reduces the loss of cooling energy.

Considering temperature rise, recovery energy, and PCM cold energy

Table 9
Cold energy consumption of the PCM.

Cases	Cold energy consumption of the PCM-kJ
Case-V0	141.71
Case-V1	158.84
Case-V2	199.90
Case-V3	286.80
Case-V2-baffle	178.91
Case-V3-baffle	215.04

consumption, Case-V2-baffle is the optimal solution. The baffle effectively controls the temperature rise of Food-2, especially in the near-door area, and prevents the rapid intrusion of hot air, thus optimizing energy efficiency and ensuring the quality of the food.

The findings of this study can improve the design of vehicle refrigeration systems by optimizing inlet velocity and baffle placement. By controlling inlet velocity and incorporating baffles, refrigerated vehicles can maintain a more stable internal temperature, reducing food temperature rise and minimizing cold energy consumption.

4. Conclusions

This study uses numerical simulation to investigate the effects of inlet velocity and baffle designs on airflow distribution, temperature fields, and energy in refrigerated vehicles. The key conclusions of this study are as follows.

- (1) When the fan is off, external hot air enters the compartment, raising the internal air temperature, and exchanges heat with the PCM and the food, resulting in a rise in food temperature. When the fan is on, the airflow from the fan collides with the incoming hot air, effectively reducing direct contact between hot air and the food. Increasing the inlet velocity further mitigates the rise in air temperature. However, when the inlet velocity reaches 3 m/s, the airflow is too large, and it directly exits the compartment, causing the internal air temperature to rise.
- (2) The baffle design effectively blocks the rapid intrusion of hot air, thereby reducing the internal air temperature rise, especially in the area between Food-2 and the door. The baffle also slows down

the temperature rise of the food. As a result, the baffle reduces the recovery energy.

- (3) As the inlet velocity increases, the cold energy consumption of the PCM significantly increases. In particular, Case-V3 exhibits the highest cold energy consumption, as the fan airflow directly carries away the PCM's cooling energy and the high inlet velocity enhances heat exchange efficiency. However, using a baffle (in Case-V2-baffle and Case-V3-baffle) reduces cold energy consumption compared to the corresponding cases without baffles.
- (4) Based on the analysis of temperature rise, recovery energy, and cold energy consumption, Case-V2-baffle is identified as the optimal solution. This scheme effectively reduces the temperature rise, ensures the quality of the food, and lowers total recovery energy. While offering the lowest recovery energy, Case-V2-baffle also performs better in terms of cold energy consumption, effectively reducing the intrusion of external hot air and maintaining a stable air temperature in the compartment.

Future research should explore the effects of PCM types and sizes on system performance. Investigating PCMs with varying melting points and sizes will provide valuable insights for optimizing the PCM-based cooling system's efficiency and improving temperature control in real-world applications.

Nomenclature

ΔH	Latent heat content (J/kg)
C_p	Specific heat J/(kg·K)
D	Hydraulic diameter of inlet (m)
d_w	Distance to the next surface (m)
F	Blending function
g	Gravity (m/s^2)
Gr	Grashof number = $g\beta_l\Delta T h_c^3/(\nu^2)$
H	Enthalpy (J/kg)
h	Sensible enthalpy (J/kg)
h_e	Convection heat transfer coefficient ($\text{W}/(\text{m}^2\cdot\text{K})$)
h_{ref}	Reference sensible enthalpy (J/kg)
h_c	Interior height of compartment (m)
k	Turbulence kinetic energy (m^2/s^2)
L	Latent heat (J/kg)
p	Pressure (Pa)
P_k	Production of $k = \min(2\nu_\epsilon S_{ij}S_{ij}, 10\beta^*k\omega)$ (m^2/s^3)
Pr	Prandtl number
Pr_t	Turbulent Prandtl number
Re	Reynolds number $u_{\text{in}}D/\nu$
S_{ij}	Mean strain rate (m/s^2)
t	Time (s)
T	Temperature (K)
T_0	Reference temperature (K)
T_{in}	Inlet air temperature (K)
T_l	Liquidus temperature (K)
T_s	Solidus temperature (K)
u_i, u_j	Velocity vector (m/s)
u_{in}	Inlet velocity (m/s)
x, y, z	x, y, z directions (m)
Y_k	Dissipation of turbulence kinetic energy = $\beta^*k\omega$ (m^2/s^3)
Y_ω	Dissipation of specific dissipation rate = $\beta\omega^2$ (s^{-2})

Greek letters

α	Model parameter
λ	Thermal conductivity $\text{W}/(\text{m}\cdot\text{K})$
α_1	Model parameter 0.55
α_2	Model parameter 0.44
β	Model parameter
β^*	Model parameter 0.09

β_1	Model parameter 0.075
β_2	Model parameter 0.0828
β_{lf}	Liquid fraction
β_t	Thermal expansion coefficient (1/K)
δ_{ij}	Kronecker delta
ν	Kinematic viscosity (m^2/s)
ν_t	Turbulent viscosity (Pa·s)
ρ	Density (kg/m^3)
σ_k	Model parameter
σ_{k1}	Model parameter 2.0
σ_{k2}	Model parameter 2.0
σ_ω	Model parameter
$\sigma_{\omega1}$	Model parameter 1.0
$\sigma_{\omega2}$	Model parameter 1.168
ω	Specific dissipation rate s^{-1}

CRediT authorship contribution statement

Puxian Ding: Writing – original draft, Validation, Methodology, Investigation, Funding acquisition. **Yingsun Sun:** Writing – review & editing, Supervision, Conceptualization. **Shiming Liu:** Project administration, Funding acquisition. **Di Fan:** Writing – review & editing, Data curation. **Ye Feng:** Visualization, Investigation. **Xinqiang Liao:** Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

Research Platforms and Projects of Universities of Department of Education of Guangdong Province (2024KTSCX319). Tertiary Education Scientific Research Project of Guangzhou Municipal Education Bureau (2024311968). Research Projects of Guangzhou Panyu Polytechnic (2024KJ07). Project supported by GDUPS (2024).

Data availability

Data will be made available on request.

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